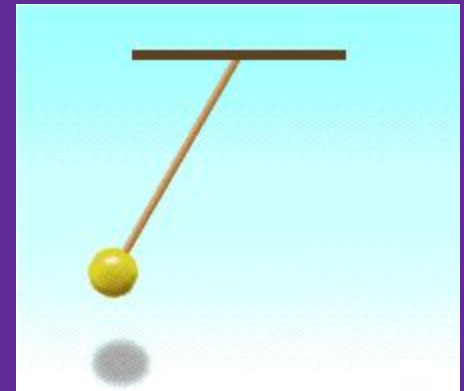


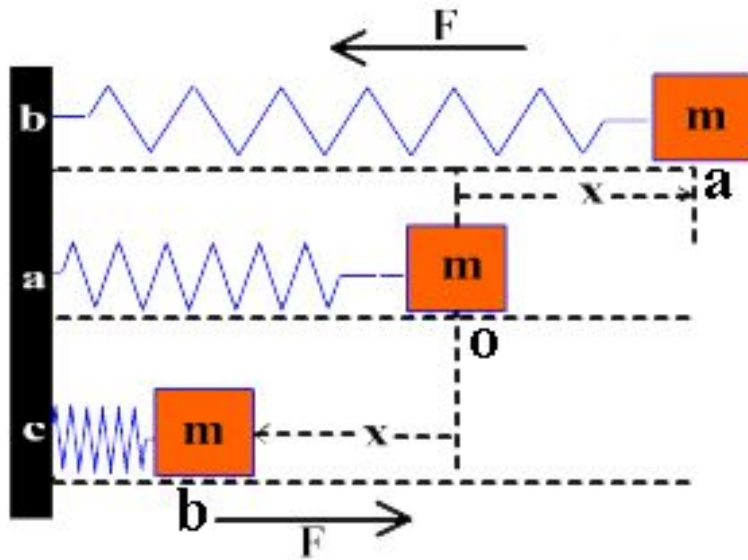
Simple Harmonic Motion

AP Physics 1



Simple Harmonic Motion

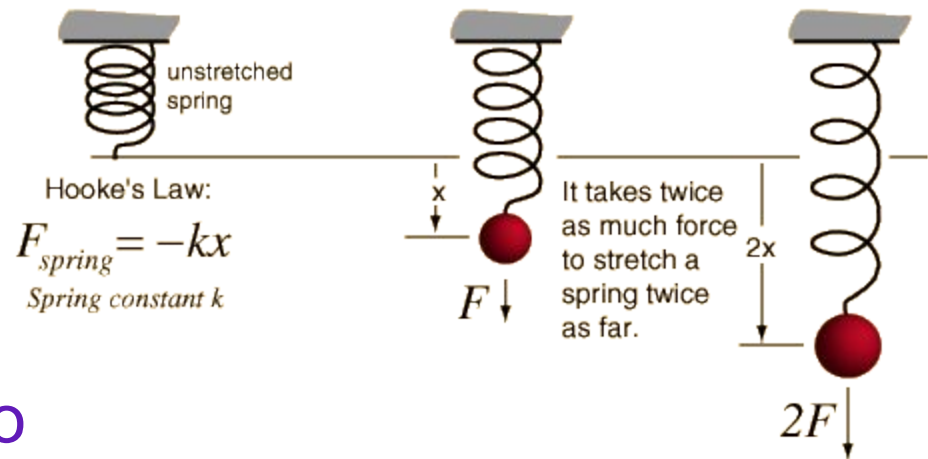
Back and forth motion that is caused by a force that is directly proportional (linear relationship) to the displacement. The displacement centers around an equilibrium position.



$$F_s \propto x$$

Springs – Hooke's Law

One of the simplest type of simple harmonic motion is called **Hooke's Law**. This is primarily in reference to **SPRINGS**.



$$F_s \propto x$$

k = Constant of Proportionality

k = Spring Constant (Unit : N/m)

$$F_s = kx \quad \text{or} \quad -kx$$

The negative sign only tells us that “F” is what is called a **RESTORING FORCE**, in that it works in the **OPPOSITE** direction of the displacement.

Hooke's Law

Common formulas which are set equal to Hooke's law are N.S.L. and weight

$$F_s = kx$$

$$F_s = F_g \rightarrow kx = mg$$

$$F_s = F_{net} \rightarrow kx = ma$$

Example

A load of 50 N attached to a spring hanging vertically stretches the spring 5.0 cm. The spring is now placed horizontally on a table and stretched 11.0 cm. What force is required to stretch the spring this amount?

$$F_s = kx$$

$$50 = k(0.05)$$

$$k = \mathbf{1000 \text{ N/m}}$$

$$F_s = kx$$

$$F_s = (1000)(0.11)$$

$$F_s = \mathbf{110 \text{ N}}$$

Hooke's Law from a Graphical Point of View

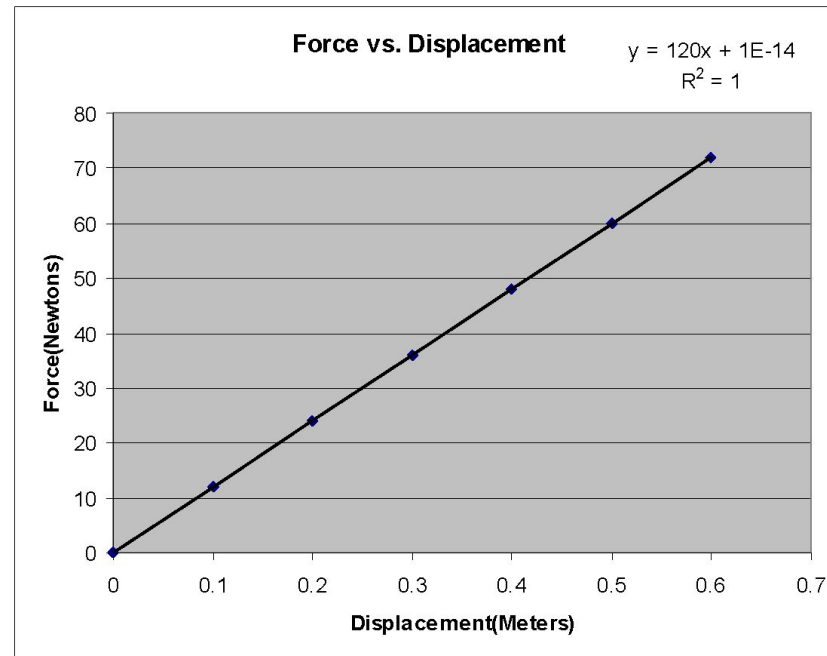
Suppose we had the following data:

x(m)	Force(N)
0	0
0.1	12
0.2	24
0.3	36
0.4	48
0.5	60
0.6	72

$$F_s = kx$$

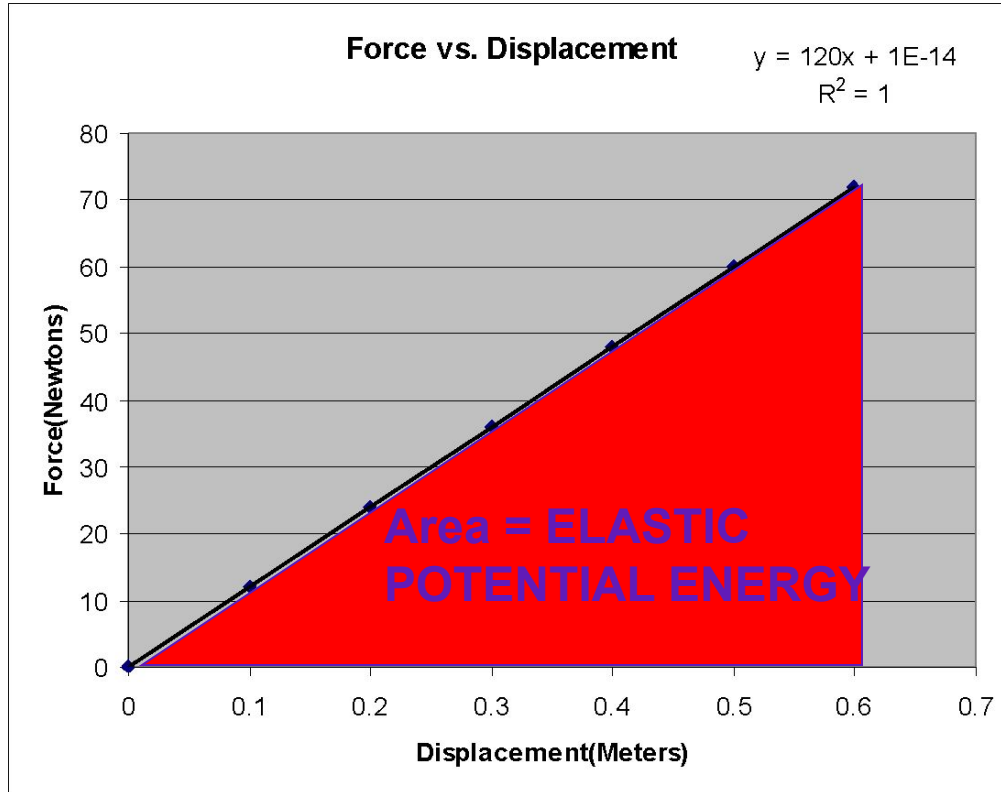
$$k = \frac{F_s}{x}$$

k = Slope of a F vs. x graph



$$k = 120 \text{ N/m}$$

We have seen F vs. x Before!!!!



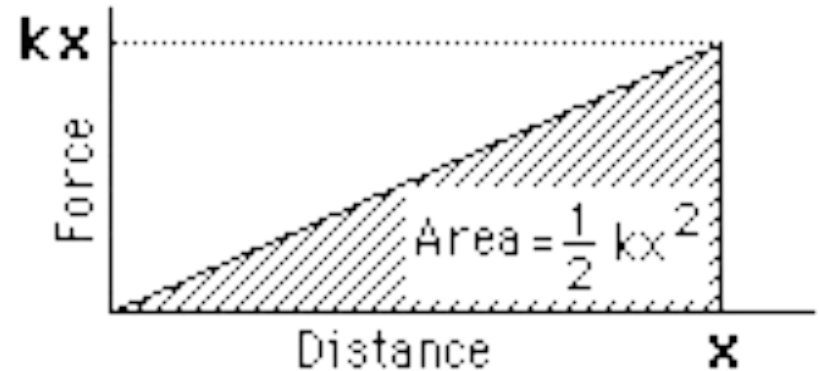
Work or ENERGY = $F\Delta x$

Since WORK or ENERGY is the AREA, we must get some type of energy when we compress or elongate the spring. This energy is the AREA under the line!

Since we STORE energy when the spring is compressed and elongated it classifies itself as a “type” of POTENTIAL ENERGY, U_s . In this case, it is called ELASTIC POTENTIAL ENERGY.

Elastic Potential Energy

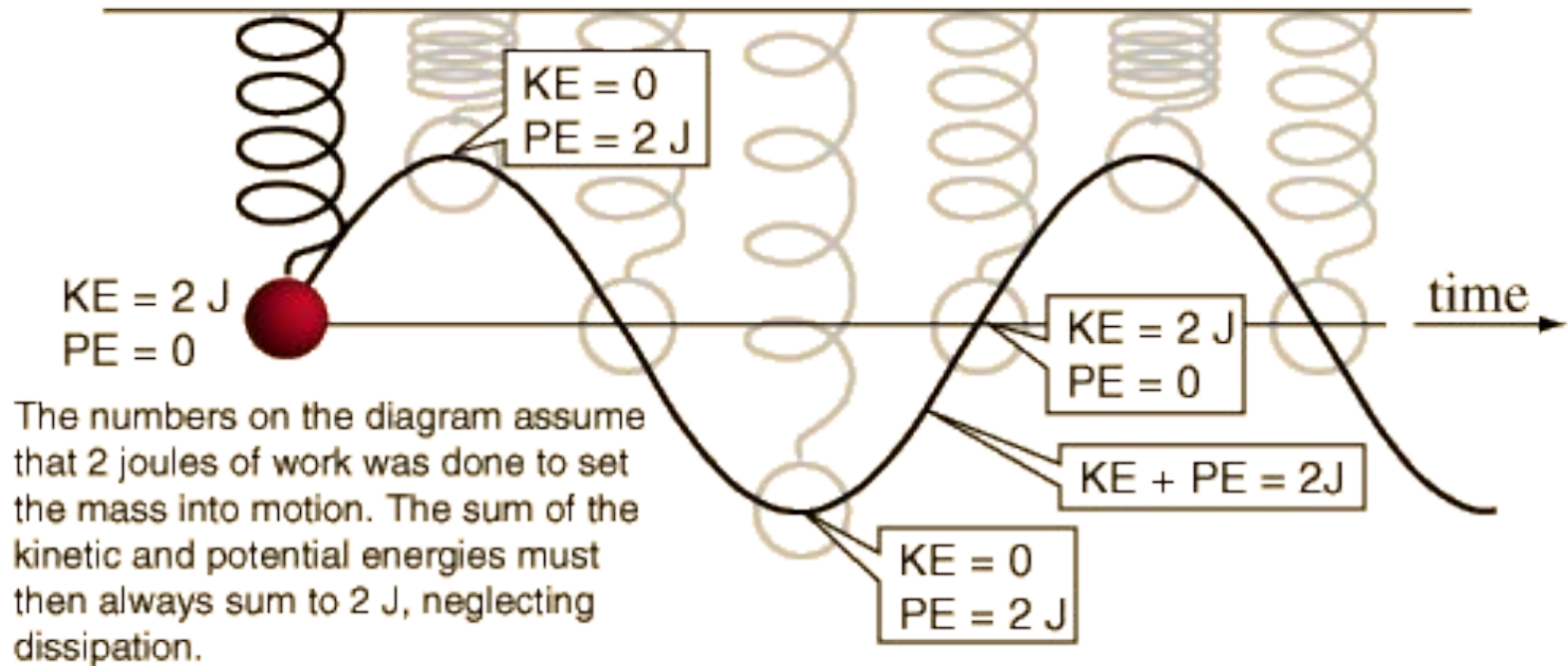
The graph of F vs. x for a spring that is IDEAL in nature will always produce a line with a positive linear slope. Thus the area under the line will always be represented as a triangle.



$$U_s = \frac{1}{2} kx^2$$

NOTE: Keep in mind that this can be applied to **WORK** or can be conserved with any other type of energy.

Conservation of Energy in Springs



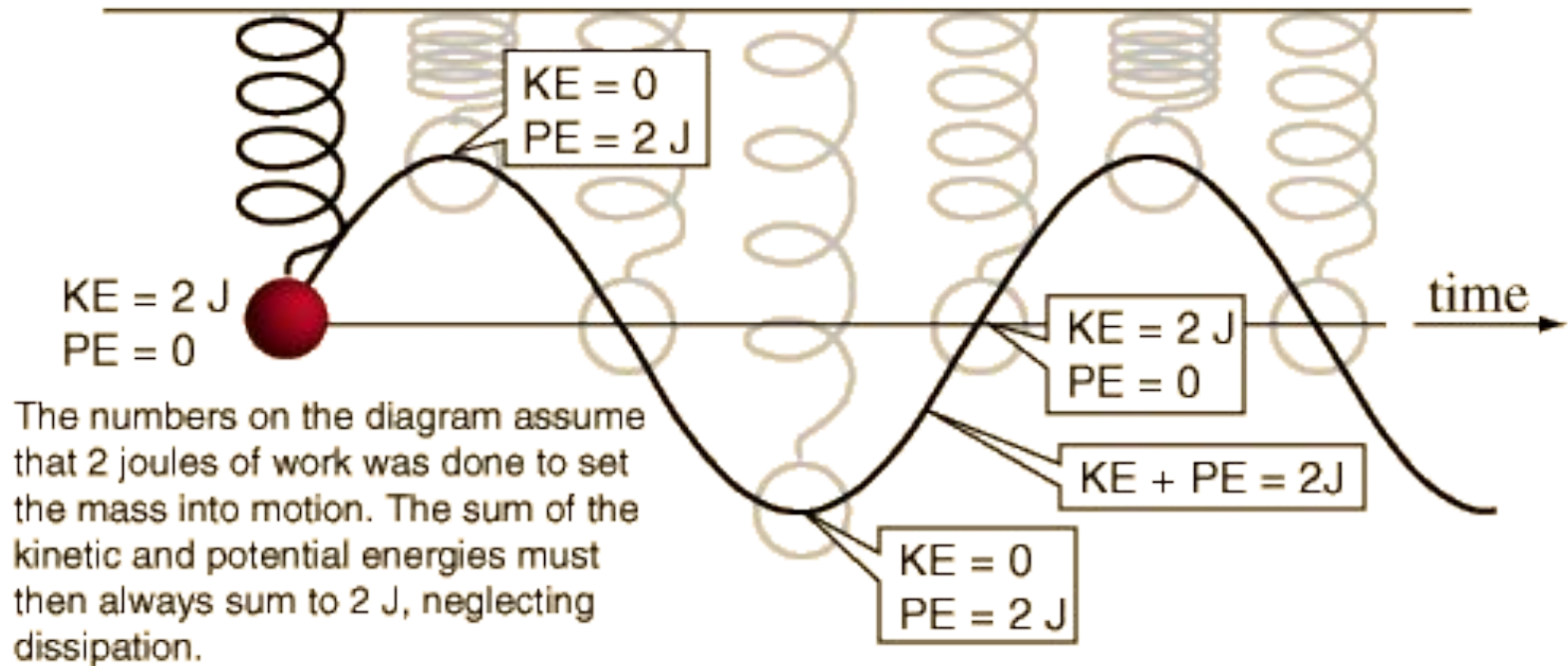
Spring energy conservation

- A 2 kg box attached to a horizontal spring ($k = 50 \text{ N/m}$) is stretched 0.5 m from equilibrium. Calculate the maximum speed of the box. The surface the box slides on is frictionless.
- The same approach can be applied to find the maximum speed of an object on a vertical spring.
- This only works for finding the maximum speed

Spring energy conservation

- A 5 kg box slides down a 10 m long frictionless incline angled at 30 degrees above the horizontal onto a frictionless level surface. It then collides with a horizontal spring ($k = 50 \text{ N/m}$). After compressing the spring, the box comes to rest. Calculate the spring compression.

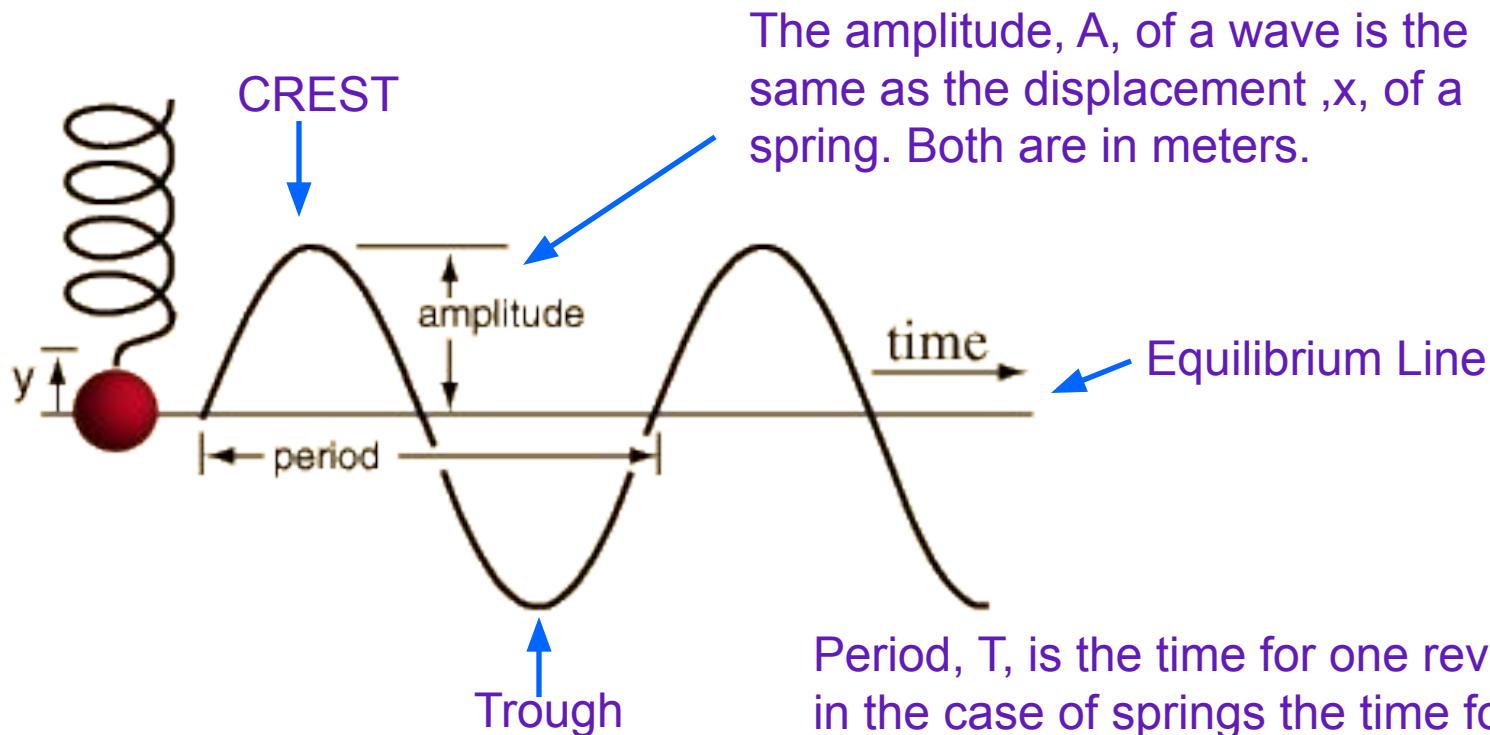
VERTICAL SPRINGS - Conservation of Energy in Springs



Spring energy

- A 4kg mass attached to a vertical spring ($k = 1000 \text{ N/m}$) is displaced 20 cm from equilibrium. Calculate its speed when it reaches equilibrium. (hint: maximum speed is achieved at equilibrium for any oscillator)
- Calculate its speed when it is 12cm from equilibrium.

Springs are like Waves and Circles



$T_s = \text{sec/cycle}$. Let's assume that the wave crosses the equilibrium line in one second intervals. $T = 3.5 \text{ seconds} / 1.75 \text{ cycles}$. **$T = 2 \text{ sec}$** .

Period, T , is the time for one revolution or in the case of springs the time for ONE COMPLETE oscillation (One crest and trough). Oscillations could also be called vibrations and **cycles**. In the wave above we have 1.75 cycles or waves or vibrations or oscillations.

Frequency

The **FREQUENCY** of a wave is the inverse of the PERIOD. That means that the frequency is the #cycles per sec. The commonly used unit is HERTZ(HZ).

$$Period = T = \frac{\text{seconds}}{\text{cycles}} = \frac{3.5s}{1.75cyc} = 2s$$

$$Frequency = f = \frac{\text{cycles}}{\text{seconds}} = \frac{1.75cyc}{3.5sec} = 0.5 \text{ c/s} = 0.5Hz$$

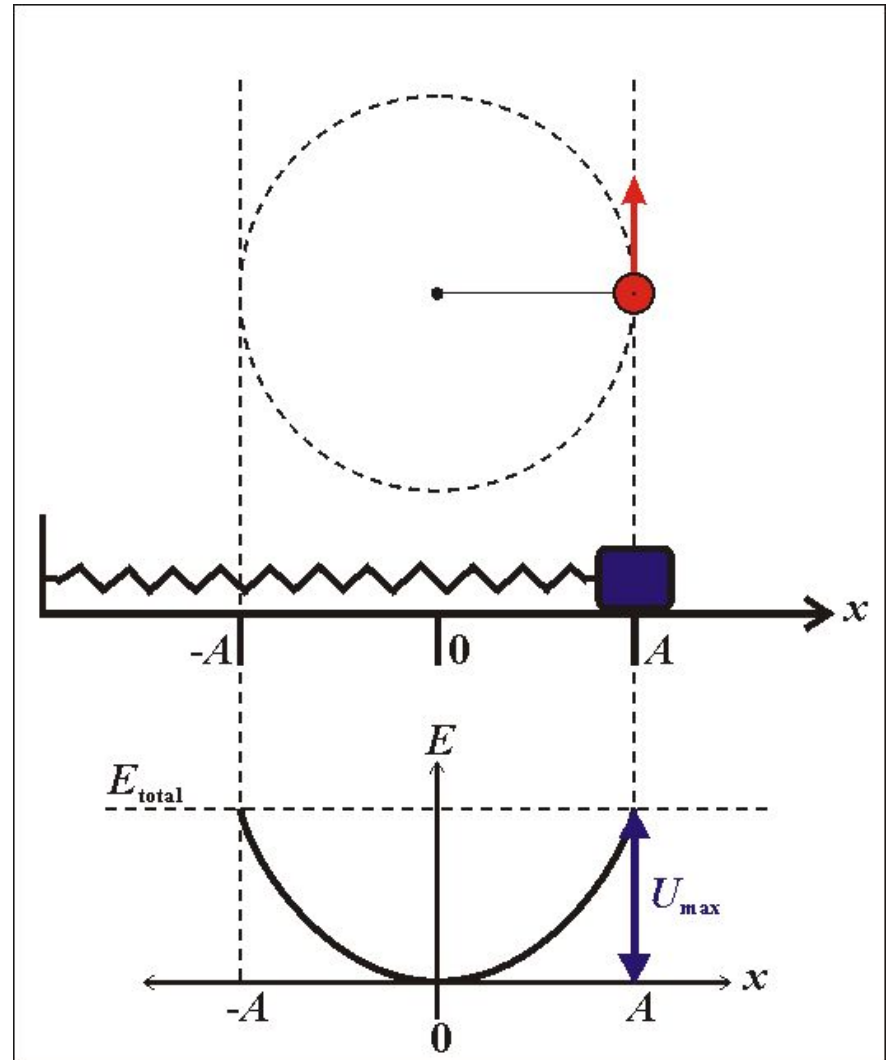
$$T = \frac{1}{f} \quad f = \frac{1}{T}$$

SHM and Uniform Circular Motion

Springs and Waves behave very similar to objects that move in circles.

The radius of the circle is symbolic of the displacement, x , of a spring or the amplitude, A , of a wave.

$$x_{spring} = A_{wave} = r_{circle}$$



SHM POSITION EQUATION

- For a system starting at equilibrium



$$x = x_0 \sin(\omega t)$$

- For a system starting at maximum displacement



$$x = x_0 \cos(\omega t)$$

$x_0 = A = \text{amplitude}$

-  Your calculator should be in radian mode

SHM and Uniform Circular Motion

$$v = \frac{2\pi r}{T}, \quad r = A$$

$$v = \frac{2\pi A}{T} \rightarrow \frac{T}{2\pi} = \frac{A}{v}$$

$$Energy_{before} = Energy_{after}$$

$$U_s = K$$

$$\frac{1}{2}kx^2 = \frac{1}{2}mv^2 \rightarrow kx^2 = mv^2, x = A$$

$$\frac{A^2}{v^2} = \frac{m}{k} \rightarrow \frac{A}{v} = \sqrt{\frac{m}{k}}$$

$$\frac{T}{2\pi} = \sqrt{\frac{m}{k}} \rightarrow T_{spring} = 2\pi \sqrt{\frac{m}{k}}$$

- The radius of a circle is symbolic of the amplitude of a wave.

- Energy is conserved as the elastic potential energy in a spring can be converted into kinetic energy. Once again the displacement of a spring is symbolic of the amplitude of a wave

- Since BOTH algebraic expressions have the ratio of the Amplitude to the velocity we can set them equal to each other.

- This derives the PERIOD of a SPRING.

Example

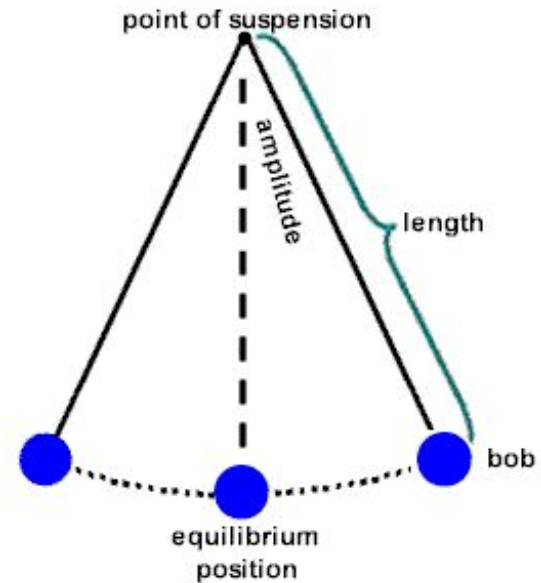
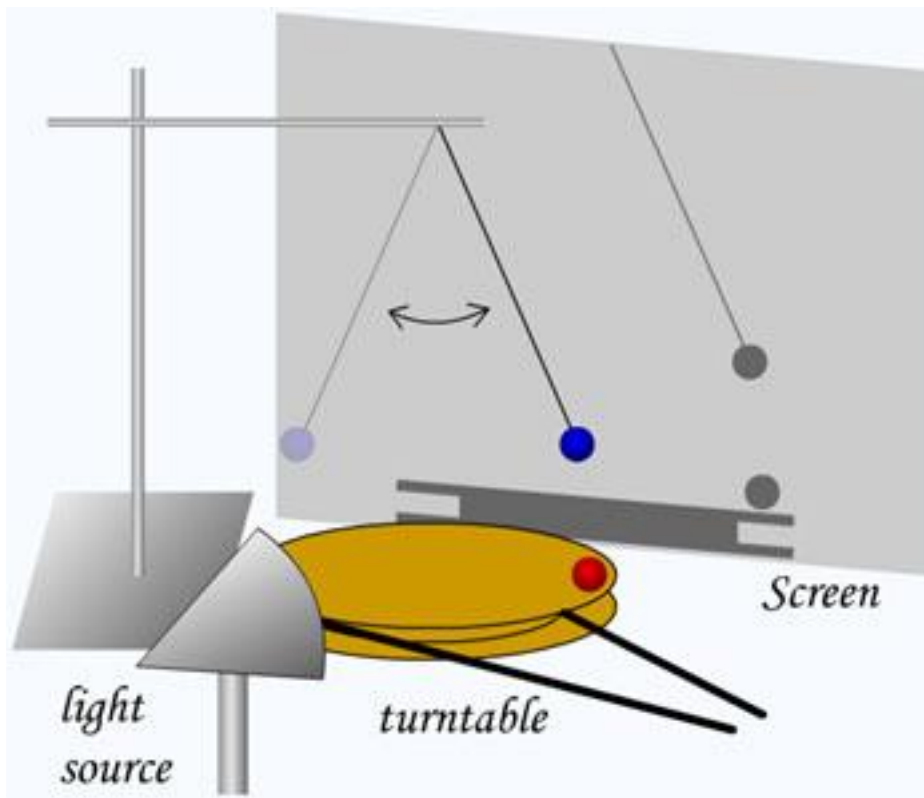
A 200 g mass is attached to a spring and executes simple harmonic motion with a period of 0.25 s. If the total energy of the system is 2.0 J, find the (a) force constant of the spring (b) the amplitude of the motion

$$T_s = 2\pi\sqrt{\frac{m}{k}} \rightarrow 0.25 = 2\pi\sqrt{\frac{0.200}{k}} \quad k = \mathbf{126.3 \text{ N/m}}$$

$$U_s = \frac{1}{2}kx^2 \rightarrow 2 = \frac{1}{2}kA^2 \quad A = \mathbf{0.18 \text{ m}}$$

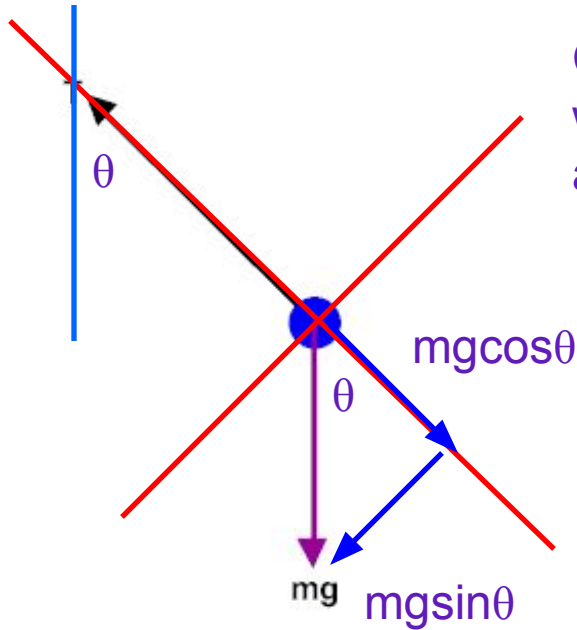
Pendulums

Pendulums, like springs, oscillate back and forth exhibiting simple harmonic behavior.



A shadow projector would show a pendulum moving in synchronization with a circle. Here, the angular amplitude is equal to the radius of a circle.

Pendulums



Consider the FBD for a pendulum. Here we have the weight and tension. Even though the weight isn't at an angle let's draw an axis along the tension.

$$mg \sin \theta = \text{Restoring Force}$$

$$mg \sin \theta = kx$$

Pendulums

$$\theta = \frac{s}{R} = \frac{s}{L}$$

$$s = \theta L = \text{Amplitude}$$

$$mg \sin \theta = k\theta L$$

$$\sin \theta \cong \theta, \text{ if } \theta = \text{small}$$

$$mg = kl$$

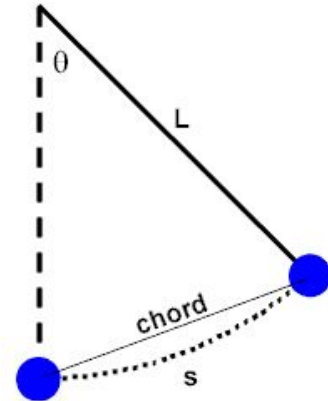
$$\frac{m}{k} = \frac{l}{g}$$

$$T_{\text{spring}} = 2\pi \sqrt{\frac{m}{k}}$$

$$mg \sin \theta = \text{Restoring Force}$$

$$mg \sin \theta = kx$$

What is x ? It is the amplitude! In the picture to the left, it represents the chord from where it was released to the bottom of the swing (equilibrium position).



$$T_{\text{pendulum}} = 2\pi \sqrt{\frac{l}{g}}$$

Example



A visitor to a lighthouse wishes to determine the height of the tower. She ties a spool of thread to a small rock to make a simple pendulum, which she hangs down the center of a spiral staircase of the tower. The period of oscillation is 9.40 s. What is the height of the tower?

$$T_P = 2\pi \sqrt{\frac{l}{g}} \rightarrow l = \text{height}$$

$$T_P^2 = \frac{4\pi^2 l}{g} \rightarrow l = \frac{T_P^2 g}{4\pi^2} = \frac{9.4^2 (9.8)}{4(3.141592)^2} =$$

$$\mathbf{L = Height = 21.93 \text{ m}}$$